



■ overlap (matched ink) ■ missing reference ink ■ extra candidate ink □ truth cells □ predicted cell

Primary cause of the 3,565 exact-region failures (diagnostic cohort: 200 references × 11 perturbations at every severity level)



Multi-region/boundary truth: 1,922 (0.539) ← case b

Local finding absent from top 3: 718 (0.201)

Wrong direction channel: 876 (0.246)

Wrong missing/extra type: 38 (0.011); Alignment residual shift: 11 (0.003)